

CleanGuard

Intelligent Restroom Environmental Monitoring Platform

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1. Project Overview

CleanGuard is an IoT-based environmental monitoring platform that protects campus and dormitory restroom facilities from freezing pipes, water leaks, and poor air quality. Battery-powered ESP8266 sensor nodes report temperature, humidity, and air-quality readings to a PHP/MySQL backend every five seconds, and a React dashboard gives facility staff real-time, color-coded visibility into every monitored restroom — replacing the periodic manual inspections that leave hours-long gaps between checks.

Capability	Description
Continuous Environmental Sensing	Every restroom reports temperature, humidity, and air quality every 5 seconds.
Real-Time Monitoring Dashboard	Facility staff view live, building- and room-level status from any browser.
Automated Status Classification	A rule-based engine flags Normal / Warning / Danger conditions automatically.
Device Management	Administrators register, edit, and remove sensor nodes (full CRUD).
Predictive Intelligence (Future Work)	ML models will forecast freeze, leak, and odor risk.

2. System Architecture

CleanGuard follows a layered architecture that separates sensing, communication, application logic, storage, and presentation. Figure 1 shows the platform's five cooperating modules: the IoT, Backend, Database, and Frontend modules are implemented and in production, while the Artificial Intelligence module is a planned extension designed to attach to the same data pipeline without requiring changes to the other layers.

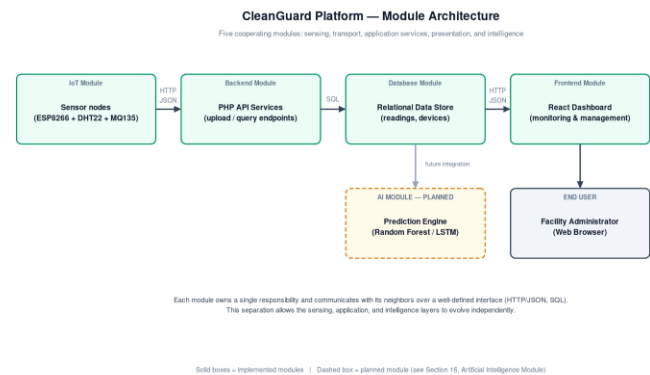


Figure 1 — CleanGuard module architecture.

Layer	Technology	Status
Sensing	ESP8266, DHT22, MQ-135	Implemented
Communication	Wi-Fi, HTTP / JSON	Implemented
Application (Backend)	PHP REST API	Implemented
Data (Database)	MySQL	Implemented
Presentation (Frontend)	React single-page application	Implemented
Intelligence (AI)		Future Work

3. Hardware & Software

Each restroom is instrumented with a single battery-capable sensor node built around a WeMos D1 Mini (ESP8266), a DHT22 temperature/humidity sensor, and an MQ-135 air-quality sensor. Figure 2 shows the first assembled prototype, used to validate wiring and firmware before field deployment.

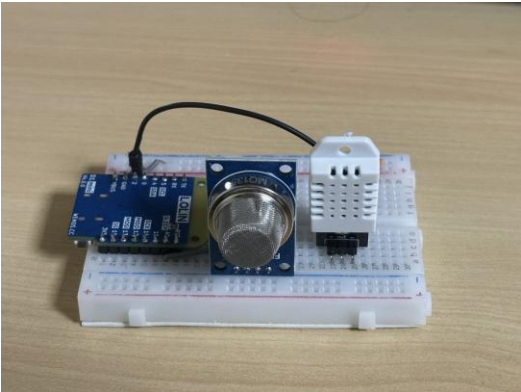


Figure 2 — First IoT sensor-node prototype (ESP8266 + DHT22 + MQ-135).

3.1 Hardware Components

Component	Role
DHT22 Sensor	Measures ambient temperature and relative humidity.
MQ-135 Sensor	Measures air quality / gas concentration.
ESP8266 (WeMos D1 Mini)	Reads sensor data and transmits it over Wi-Fi.
Li-ion Battery + TP4056	Powers the sensor node independent of fixed wiring.

3.2 Software Stack

Layer	Technologies
Frontend	React 19, Vite, Tailwind CSS, React Router, Recharts
Backend	PHP REST endpoints, MySQL
IoT Firmware	Arduino / C++ on ESP8266
Planned AI	

4. Dashboard & AI Features

The dashboard supports three modes: Live and Test show the latest reading from every device, refreshed automatically every few seconds; History provides a paginated, chronological log of every measurement stored in the database, with automatic refresh disabled for stable browsing. A Building Selector lets staff filter the monitored bathrooms by building. The interface is fully responsive, adapting the same layout, colors, and components across desktop, tablet, and mobile screens.

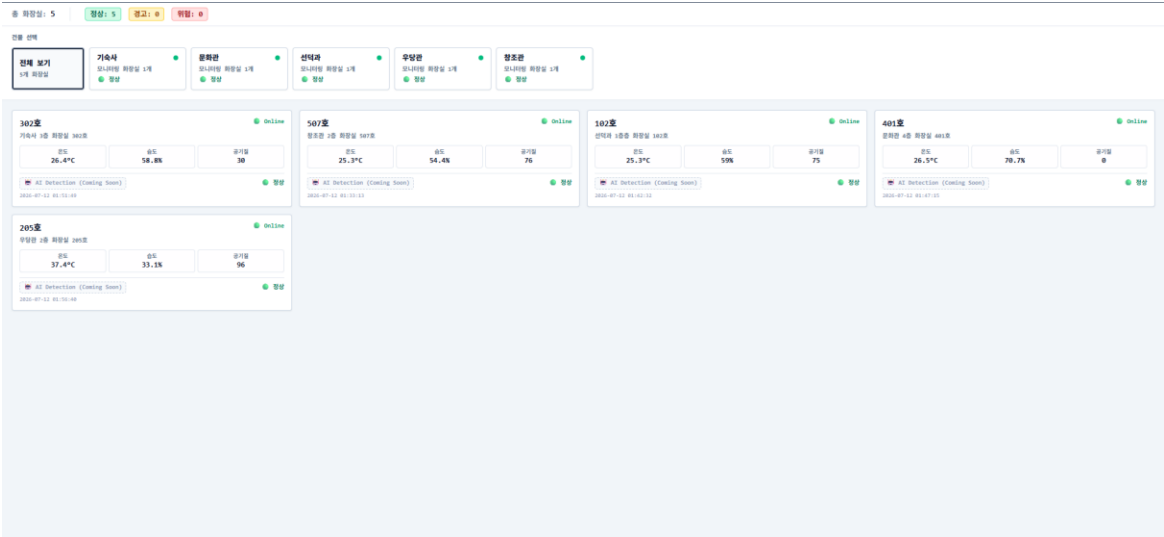


Figure 3 — Main monitoring dashboard: building selection and real-time bathroom status.

Selecting a bathroom opens a detail panel (Figure 4) with the device's identification, current readings, and a historical trend chart. The panel also reserves a clearly labeled placeholder for AI-based window-state detection, so the interface is ready for the feature without any redesign once it ships.

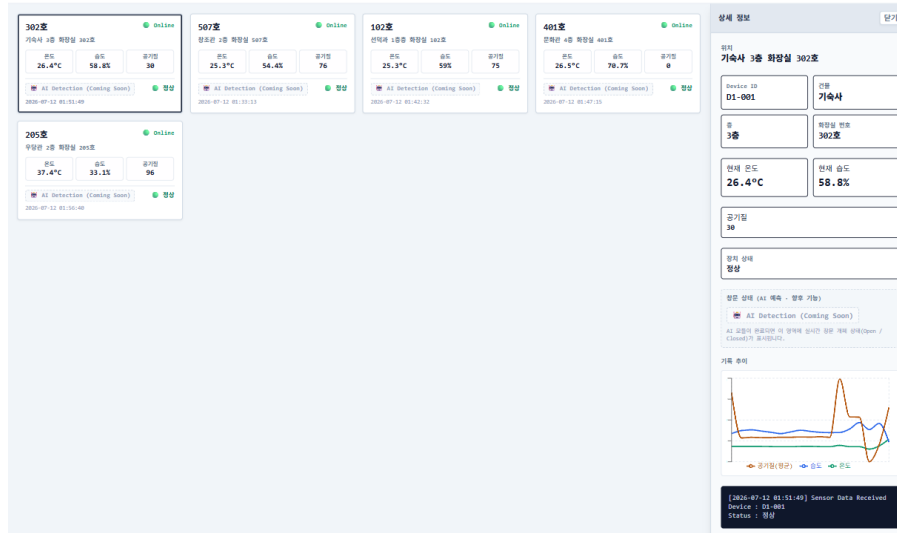


Figure 4 — Detail panel with historical trend chart. "AI Detection (Coming Soon)" marks a Future Work item, not an implemented feature.

A separate Device Management screen (Figure 5) gives administrators full create / edit / delete control over the sensor-node registry, with each device's building, floor, and bathroom number editable independently of its permanent device ID.

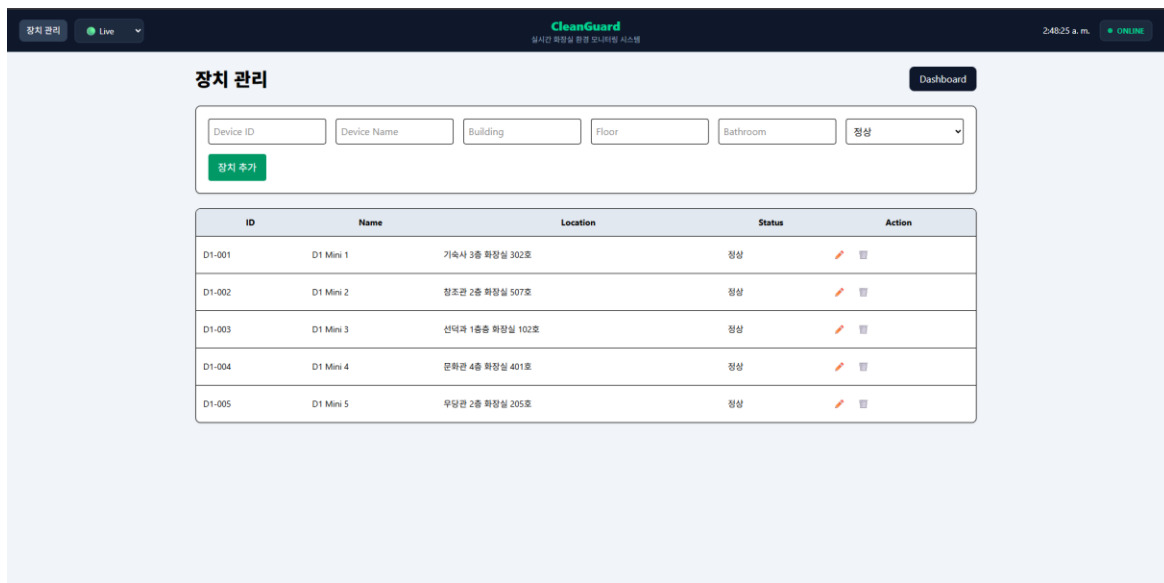


Figure 5 — Device Management screen: full CRUD control over the sensor-node registry.

Future Work — AI Window Detection

- Freeze-risk prediction using LSTM time-series forecasting.
- Leak prediction using Random Forest classification.
- Odor classification using Random Forest / gradient-boosted models.

These models are designed but not yet trained or deployed. No AI inference currently runs anywhere in the platform; every reading shown today is either a raw sensor value or the output of the deterministic status-classification rules described in Section 2.

5. Results & Conclusion

CleanGuard delivers a working, end-to-end restroom environmental-monitoring platform: sensor nodes report readings on a fixed interval, a PHP/MySQL backend stores and serves them, and a responsive React dashboard gives administrators real-time visibility, historical browsing, and full device management. Its modular architecture was built so the planned AI features — freeze, leak, and odor detection — can be added as a direct extension of the existing data pipeline, without further redesign.